

46-921: Financial Data Science I

Fall 2020

Course Description

The first in a two-course sequence covering methods of extracting useful information from raw financial data. Focus is placed on tools of data exploration and mining, with particular emphasis on handling large and complex financial data sets. Topics include data cleaning, data visualization, natural language processing, nonparametric smoothing techniques, and unsupervised learning.

Prerequisites

There are no formal course prerequisites. It is assumed that you have taken a rigorous, University-level probability course. It is assumed that you understand basic matrix algebra operations. Finally, a basic understanding of the fundamentals of computer programming is important.

Learning Objectives

After this course, students will be able to do the following:

1. Appreciate the importance of visualization as a tool of understanding the data, developing hypotheses, and determining appropriate models. Be able to implement a wide range of visualization strategies depending on the needs of the situation, and the type of data.
2. Be aware of basic issues encountered when working with real data, including missing data and processing data of different types.
3. Understand the theory and application of simple linear regression.
4. Understand how to use nonparametric density estimation and smoothing to fit to data with minimal assumptions.
5. Know how to implement methods of unsupervised learning such as principal components analysis, nonlinear dimension reduction, and clustering, and understand how these approaches can yield useful representations and useful insight.

Lecturer

Chad M. Schafer, Department of Statistics & Data Science

cschafer@cmu.edu

Lecture Schedule

Lectures will be from 9:50 AM to 11:20 AM EDT (UTC-4) on Tuesdays and Thursdays, starting on September 1 and ending on October 15.

Reference Books

The following books are useful, and in some cases references will be made to them during the course. It is not necessary that you purchase these, however.

- *Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython*, First Edition, by Wes McKinney. 2017. O'Reilly Media. ISBN: 978-1491957660.

- *Statistics and Data Analysis for Financial Engineering: with R examples*, Second Edition, by David Ruppert and David Matteson. 2015. Springer. ISBN: 978-1493926138. Available at <https://link.springer.com/book/10.1007/978-1-4939-2614-5>

- *Introduction to Statistical Learning* by Gareth James, Daniela Witten, Trevor Hastie and Robert Tibshirani. 2013. Springer. ISBN: 978-1461471370. Available at <https://faculty.marshall.usc.edu/gareth-james/ISL/>.

- *Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems*, First Edition, by Aurelien Geron. 2017. O'Reilly Media. ISBN: 978-1491962299.

Lecture Delivery

Lectures will be delivered via Zoom. You can access them directly via this URL:

<https://cmu.zoom.us/j/96791922778?pwd=cnBzN25ydWxxVjRCZjZrWXZOaGRhdz09.>

The Meeting ID is 967 9192 2778 and the passcode is 211692.

The lecture notes are available in the "Pre-Lecture Notes" folder on the "Files" page.

The lecture notes will be delivered via the document camera. These will be scanned into a .pdf file and posted in the "Post-Lecture Notes" folder on the "Files" page following the lecture.

You should write the notes yourself during the lecture. In my opinion, there is no substitute for this in the learning process.

You are encouraged to ask questions at any time during the lectures. If I feel that your question ventures into technical details which are not pertinent to the discussion, I may ask you to either come to office hours or send me an email. You should not interpret this as me discouraging questions and discussion.

Lectures will be recorded and posted here.

Computing

Instruction in this course will include learning how to use Python. Some lectures, homeworks, and the final project will require the use of Jupyter notebooks, so installing the necessary software is important.

Identification within Zoom

Please ensure that your name as shown on Zoom matches that as listed on the Canvas "People" page. Also, you should include preferred pronouns along with your name on Zoom. (You can add them as part of your "Last Name" by editing your profile on Zoom.)

Asking Questions

Times for Zoom "office hours" will be arranged and announced. In general, I am very flexible and will make myself available for questions as needed. We will also use the Discussions feature of Canvas to handle questions.

Email

Please do not email TAs with concerns over grading, or any other issues with the course. Instead, email the instructor.

I assume that you read, at least daily, your official University email (your @andrew.cmu.edu address). Email can be used for important announcements, so make sure that you are regularly reading mail sent this account.

You can email me any questions you have regarding the lecture and/or homework. If I feel it is appropriate, I will ask you to post your question to Canvas so that everyone can read it. Ultimately, it will be quicker if you post the question directly on Canvas; I will still see it immediately.

I am good about responding to email, and try to be helpful. However, you should be reasonable about expectations of responses to questions close to when homeworks are due. The time leading up a lecture tends to be the busiest time of the week for me.

Course Grading

Grading will be based on the five homework assignments (50%) and the final project (50%).

Homeworks

There will be five homework assignments, due at 9:30 AM EDT on Thursdays: September 10, 17, 24, October 1, and 8.

Homeworks will be submitted as a pdf file, via Canvas. In some cases you will also be asked to submit a Jupyter notebook.

Final Project

There will not be a final exam in this course, but instead a final project. Students will be divided into groups **by the instructor** for the purposes of this project. All work on the final project will have concluded on or before October 15, i.e., prior to the start of the final exam period for Mini 1.

More details regarding the final project will be made available soon.

Integrity

Please read CMU's [Policy on Academic Integrity](#) and the "Code of Conduct" section of the MSCF Handbook.

You are not allowed to utilize any materials from past versions of this course, unless I provide them to you. Any such use will be considered cheating, and treated as such.

Collaboration is a tricky issue. The support and assistance of your classmates can go a long way towards helping you to understand the material. Ultimately, however, you are responsible for preparing yourself for the final exam, and for the later courses that build on probability. I encourage you to collaborate, but, when it comes time to write up your solutions, you should work independently and each student should make sure that he/she understands the results. Instances where students copy the work of another student will be treated as cheating.

This course will include group work. **Students must remember that they are responsible for all material that is handed in or presented with their name attached.** Dividing the work of the project can be a good strategy, but it does absolve group members from responsibility for academic integrity violations that may be present in another student's contribution.

Diversity within our Class, CMU, and Beyond

We must treat every individual with respect. We are diverse in many ways, and this diversity is fundamental to building and maintaining an equitable and inclusive campus community.

Diversity can refer to multiple ways that we identify ourselves, including but not limited to race, color, national origin, language, sex, disability, age, sexual orientation, gender identity, religion, creed, ancestry, belief, veteran status, or genetic information. Each of these diverse identities, along with many others not mentioned here, shape the perspectives our students, faculty, and staff bring to our campus. We, at CMU, will work to promote diversity, equity and inclusion not only because diversity fuels excellence and innovation, but because we want to pursue justice. We acknowledge our imperfections while we also fully commit to the work, inside and outside of

our classrooms, of building and sustaining a campus community that increasingly embraces these core values.

Each of us is responsible for creating a safer, more inclusive environment.

Unfortunately, incidents of bias or discrimination do occur, whether intentional or unintentional. They contribute to creating an unwelcoming environment for individuals and groups at the university. Therefore, the university encourages anyone who experiences or observes unfair or hostile treatment on the basis of identity to speak out for justice and support, within the moment of the incident or after the incident has passed. Anyone can share these experiences using the following resources:

- **Center for Student Diversity and Inclusion:** csdi@andrew.cmu.edu, (412) 268-2150
- **Report-It online anonymous reporting platform:** reportit.net username: *tartans* password: *plaid*

All reports will be documented and deliberated to determine if there should be any following actions. Regardless of incident type, the university will use all shared experiences to transform our campus climate to be more equitable and just.

Accommodations for Students with Disabilities

If you have a disability and have an accommodations letter from the Disability Resources office, I encourage you to discuss your accommodations and needs with me as early in the course as possible. I will work with you to ensure that accommodations are provided as appropriate. If you suspect that you may have a disability and would benefit from accommodations but are not yet registered with the [Office of Disability Resources](#), I encourage you to contact them at access@andrew.cmu.edu.

Your Well-Being

Take care of yourself. Do your best to maintain a healthy lifestyle this semester by eating well, exercising, avoiding drugs and alcohol, getting enough sleep and taking some time to relax. This will help you achieve your goals and cope with stress.

All of us benefit from support during times of struggle. There are many helpful resources available on campus and an important part of the college experience is learning how to ask for help. Asking for support sooner rather than later is almost always helpful.

If you or anyone you know experiences any academic stress, difficult life events, or feelings like anxiety or depression, we strongly encourage you to seek support. Counseling and

Psychological Services (CaPS) is here to help: call 412-268-2922 and visit their website at <http://www.cmu.edu/counseling/>. Consider reaching out to a friend, faculty or family member you trust for help getting connected to the support that can help.

If you or someone you know is feeling suicidal or in danger of self-harm, call someone immediately, day or night:

CaPS: 412-268-2922

Re:solve Crisis Network: 888-796-8226

If the situation is life threatening, call the police

On campus: CMU Police: 412-268-2323

Off campus: 911